CHANGYUE (WILLIAM) ZHAO

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EDUCATION

Duke University, Durham, NC *08/2025 – Present*
Master of Biostatistics, Health AI Track
- GPA: 4.0/4.0
- *Selected Coursework: Statistical Methods for Learning and Discovery, Deep Learning for Health Data, Natural Language Processing, High-Dimensional Statistics and Machine Learning, Real Analysis (Instructor: Michael C. Reed, ranked 2nd), Statistical Inference (Ph.D. Level)*

University of Minnesota, Twin Cities, Minneapolis, MN *09/2023 – 05/2025*
Bachelor of Science in Statistical Science, Minor in Computer Science
- GPA: 4.0/4.0
- *Selected Coursework: Mathematical Foundation (Introduction to Mathematical Analysis), Regression and Correlation Data, Statistical Computing in R, Theory of Statistics, Statistical Machine Learning, Analysis of Categorical Data, Applied Multivariate Methods, Discrete Mathematics, Advanced Programming Principles, Algorithms and Data Structures, Applied Machine Learning, Data Mining (Graduate Level)*
- Graduated with High Distinction.
- Received the Buehler Memorial Undergraduate Award as one of eight nominees.
- Named to the CLA Dean's List in all semesters (4/4).

Shandong University of Finance and Economics, Jinan, China *09/2021 – 06/2023*
Bachelor of Economics in Finance
- GPA: 92.86/100 (WES GPA: 3.89/4.0)
- *Selected Coursework: Advanced Mathematics (Calculus, Introduction to Ordinary Differential Equations), Linear Algebra, Probability Theory and Statistics, Applied Statistics, Python*
- Received the SDUFE First-Prize Scholarship in 2022 and 2023.

RESEARCH INTERESTS

Methods and systems for foundation models (particularly LLMs) and AI agents
Applications of foundation models in biomedical informatics
LLM-based reasoning and workflow automation for single-cell analysis
Deep learning methods for biomedical data analysis and multimodal integration

PUBLICATIONS

- **Zhao, C.**, and Ji, Z. (2026). Visual LLM-guided consensus spatial domain detection with L-STAR. *Nature Communications*. doi:10.1038/s41467-026-78923-w.

RESEARCH EXPERIENCE

TRACE: Trajectory Inference via AI-guided Cross-view Evidence *01/2026 – Present*
Department of Biostatistics and Bioinformatics, Duke University, Durham, NC
- Develop TRACE, an LLM-guided trajectory inference agent that automatically identifies root states and achieves high agreement with reference trajectories across evaluated scRNA-seq, scATAC-seq, and spatial transcriptomics datasets.
- Build `trace`, an end-to-end Python package using LangGraph to coordinate LLM reasoning with R analysis. Integrate Seurat preprocessing, dimensionality reduction, clustering, and marker analysis with Signac gene-activity estimation.
- Design structured prompts and targeted molecular-evidence queries to reconcile UMAP and t-SNE views for cell-type annotation, lineage reconstruction, and root-state selection.
- Develop spatial pseudotime inference using molecular evidence and spatial neighborhood graphs to distinguish repeated local processes. Estimate pseudotime from local root cells while retaining ambiguous assignments and partially resolved trajectories.
- Design `pPOS` to jointly evaluate lineage membership and within-lineage ordering, using blockwise pair counting and inclusion-exclusion to avoid double-counting cell pairs shared across branches.

- Benchmark TRACE against Monocle 3, Slingshot, and TSCAN. Investigate discrepancies in `dyneval`-based `F1branches` evaluation and reimplement the published definition over the full cell set, with diagnostics for cell coverage and branch recovery.
- **Advisor:** Prof. Zhicheng Ji, Duke University

Med2State *11/2025 – Present*

School of Medicine, Duke University, Durham, NC

- Develop Med2State, a Python/PyTorch deep learning model for structured patient-state representations from incomplete multimodal records, with separate disease, procedure, and treatment channels.
- Build modular encoders for demographics, measurements, structured EHR events, text, and images. Explore semantic cross-attention and low-rank fusion to integrate available evidence.
- Explore temporal models for irregular visits and develop learning objectives and clinical concept readouts for longitudinal patient-state representations.
- **Advisor:** Prof. Anru Zhang, Duke University

L-STAR: Visual LLM-Guided Spatial Domain Detection *Sep 2025 – Oct 2026*

Department of Biostatistics and Bioinformatics, Duke University, Durham, NC

- Developed L-STAR, a visual LLM-guided spatial domain detection framework that achieved the highest average normalized ARI across six spatial transcriptomics datasets, outperforming individual methods and fixed-method consensus.
- Built `lstar`, a Python package integrating spatial map visualization, LLM-guided method selection, consensus clustering, and biological domain annotation with Scanpy-based marker analysis.
- Designed structured prompts for repeated pairwise comparison of domain maps using tissue context and optional histology. Implemented win-rate ranking and consensus clustering in R using co-assignment matrices.
- Processed study datasets with Seurat and SpatialExperiment and developed marker-guided domain annotation, distinguishing tissue regions from cell types by sampling level and retaining `Unknown` for insufficient evidence. Evaluated annotation agreement using BioLORD-2023 embeddings and within-dataset permutation tests.
- Assessed sensitivity to ensemble size, LLM choice, color palettes, and presentation order, and quantified redundancy among selected methods.
- **Open-source software:** L-STAR **Advisor:** Prof. Zhicheng Ji, Duke University

PaperEval *03/2025 – 09/2025*

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Built resumable OpenReview pipelines with Selenium, asynchronous requests, and structured JSON storage for papers, reviews, scores, and responses. Enriched submission metadata by extracting author contact information from PDFs with PyMuPDF.
- Fine-tuned GPT-4o for paper acceptance prediction and evaluated performance with progressively richer inputs, from paper metadata to reviewer scores and review text.
- Developed LLM-based agents to assess figures and tables in the context of full manuscripts and suggest improvements to clarity, readability, and presentation.
- Fine-tuned GPT-4o mini as a grader for author responses, constructing proxy scoring targets from acceptance levels with bounded Gaussian noise.
- Trained a reply-generation model and evaluated its responses against author replies from accepted and rejected submissions using the fine-tuned grader.
- **Open-source software:** PaperEval **Advisor:** Prof. Jie Ding, University of Minnesota

MUDAC (Midwest Undergraduate Data Analysis Competition) 2025 *03/2025 – 04/2025*

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Engineered features from sparse, high-dimensional numerical and text data for mentoring-match prediction, including embeddings, sentiment scores, and pairwise similarities.
- Modeled match duration and closure using Cox proportional hazards regression and a causal Bayesian network.
- Designed a mixture-of-experts model with regression, MLP, and Transformer experts for different feature types and a learned gating network. Compared it with random forest and XGBoost baselines.
- Placed 7th in Innovation & Creativity among approximately 200 undergraduate teams at MUDAC 2025.
- **Advisors:** Barbara Kuzmak and Robert Oliver VandenBurg, University of Minnesota

UMN & Hennepin County Public Health Equity Data Challenge *10/2024 – 11/2024*
School of Public Health, University of Minnesota, Twin Cities, Minneapolis, MN

- Analyzed inequities in access to mental-health resources using Hennepin County questionnaire data, applying multiple regression, ridge regression, decision trees, and causal inference methods.
- Developed a `tkinter` interface for mental-health prediction using AdaBoost in scikit-learn.
- Co-authored the analysis report and poster and placed in the top 25% of the undergraduate division.
- **Advisors:** Marta Shore and Prof. Julian Wolfson, University of Minnesota

Overlapping Group LASSO Regression R Package Implementation *09/2024 – 05/2025*
School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Compared `overlapping_LeastR` from the MATLAB SLEP library with `oglasso` in the R package VennLasso under matched problem sizes and convergence tolerances.
- Designed a simulation study of 1,000 large sparse-matrix datasets with overlapping groups and used t-tests to compare implementation runtimes.
- Implemented the optimized overlapping-group LASSO algorithm in C for integration with R.
- **Advisor:** Prof. Tianxi Li, University of Minnesota

PROFESSIONAL EXPERIENCE

Teaching Assistant: BIOSTAT 830 *08/2026 – Present*
Department of Biostatistics and Bioinformatics, Duke University, Durham, NC

- Help build BIOSTAT 830 resources from the ground up and quickly adapt lecture materials for its first offering in Fall 2026, bridging undergraduate statistical theory and PhD-level inference.
- Hold weekly office hours and grade assignments for a class of 22 students in statistical inference, asymptotic theory, and decision theory. Discuss common errors with the instructor to guide future lectures.
- Prepare and verify proof-based LaTeX solutions, format and proofread slides, and manage materials in Canvas. Standardize notation and convert formulas into editable PowerPoint equations.
- **Instructor:** Dr. Lin (Lynn) Lin, Duke University

Undergraduate Teaching Assistant: STAT 3011 *09/2024 – 05/2025*
School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Taught weekly 50-minute sections for about 30 students on probability, sampling, hypothesis testing, regression, and statistical programming in R.
- Prepared instructional slides and managed course materials in Canvas for the teaching team. Held office hours and graded weekly assignments and exam responses through Gradescope.
- English-language teaching qualification: ELP 1, the highest rating on the Spoken English Test for Teaching Assistants (equivalent to TOEFL iBT Speaking 27–30/30 or IELTS Speaking 8.5–9.0/9).
- **Course materials:** GitHub repository
- **Instructors:** Yuyoung Park, Aritra Banerjee, Kazeem Adepoju, University of Minnesota

OTHER INFORMATION

Programming	Python, R, C/C++, Java, OCaml, Swift, MATLAB, SQL
Machine Learning	PyTorch, TensorFlow, scikit-learn, Hugging Face, NumPy, pandas
Bioinformatics	Seurat, Signac, Scanpy, scDesign3
Tools	Git, SQLite, Selenium, LangGraph, LaTeX, Markdown, SPSS
Hobbies	Photography, Snowboarding, Car Racing, Cycling, Fitness Exercise